

Generation Interconnection System Impact Study Report

For Queue Project Q17-0818

Grove Street 138 kV

July 28, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0818. The Interconnection Customer proposes a 256.7 MW battery energy storage facility interconnecting at the Grove Street 138 kV substation in the Virginia area. The request was received on June 28, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 244 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 8 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$34,428,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
AMBER-BREMO BLUFF 138 KV	BLUFF CITY-LAKE LURE 138 KV	111.2	194	18.7
BREMO BLUFF SUBSTATION-BLUFF PT WIND FACILITY 138 KV Line	LOUISA GENERATION FACILITY-FLUVANNA P.S 345 KV Line	107.8	328	8.7
MCAULEY-VIDEO PD 138 KV Line	URBAN-AMERICAN 345 KV	104.8	228	5.1
RICHLAND-DEKALB 500 KV	SAN VLY-TREGO 500 KV	104.5	2,987	9.6
HINTON AP-LOWMOOR 345 KV	MCCLAIN ENERGY FACILITY-VASEAUX LAKE 138 KV	103.0	1,182	8.1
DECLO-NEW LONDON SUBSTATION 138 KV Line	AMERICAN-URBAN 345 KV line	102.5	425	5.6
MUSKOGEE SUBSTATION-SWAN VLY 500 KV L	RICHLAND-DEKALB 500 KV L	101.9	2,943	5.3
TREGO-SWAN VLY 500 KV	TROY ENERGY LLC-RICHLAND 500 KV	101.5	3,663	5.1

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
AMBER-BREMO BLUFF 138 KV	Rebuild line with higher-capacity conductor	\$11,313,000	50% (\$5,660,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
BLUFF PT WIND FACILITY-BREMO BLUFF 138 KV	Reconductor line	\$4,012,000	36% (\$1,465,000)
MCAULEY-VIDEO PD 138 KV Line	Reconductor line	\$1,156,000	23% (\$270,000)
RICHLAND-DEKALB 500 KV Line	Reconductor line	\$75,908,000	3% (\$1,999,000)
HINTON AP-LOWMOOR 345 KV	Reconductor line	\$50,797,000	3% (\$1,597,000)
DECLO-NEW LONDON SUBSTATION 138 KV Line	Reconductor line	\$9,412,000	5% (\$479,000)
MUSKOGEE SUBSTATION-SWAN VLY 500 KV L	Reconductor line	\$48,749,000	9% (\$4,464,000)
TREGO-SWAN VLY 500 KV	Reconductor line	\$53,549,000	34% (\$18,494,000)
	Total allocated to project		\$34,428,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.